GCSE PHYSICS

Higher Tier – Paper 2 style

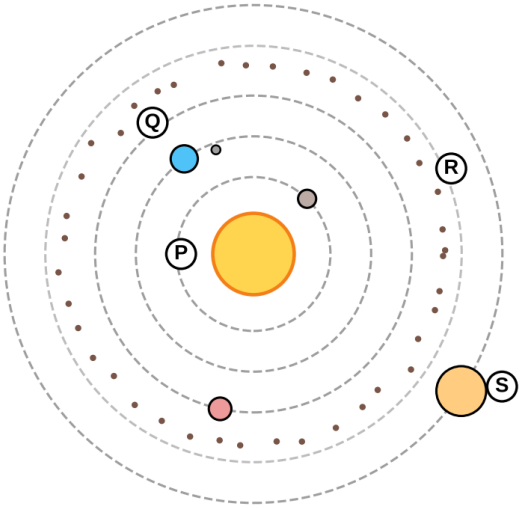
Topic 4.8: Space physics

Practice questions in the style of AQA GCSE Physics (8463). Every question is original; the layout, command words, mark allocations and mark scheme follow the real exam format.

Time allowed	About 105 minutes (roughly 1 minute per mark)
Materials	For this paper you must have: a ruler, a pencil and a calculator.
Instructions	• Use black ink or a black ball-point pen (pencil for diagrams and graphs). • Answer **all** questions. • Answer the questions in the spaces provided. • Do all rough work on a separate sheet. • Cross through any work you do not want to be marked.
Information	• The maximum mark for this paper is **99**. • The marks for questions are shown in brackets. • You are allowed to use a calculator. • You are reminded of the need for good English and clear presentation in your answers. • Questions that ask you to *describe* or *explain* are marked for the science you include, so give each point its own sentence.

Name: _____ Date: _____ Mark: _____ / 99

0 1 The diagram shows part of the Solar System. It is not to scale.



0 1 . 1 Name the objects labelled **P**, **Q**, **R** and **S**. **[4 marks]**

0 1 . 2 The Solar System is in a galaxy. Name the galaxy. **[1 mark]**

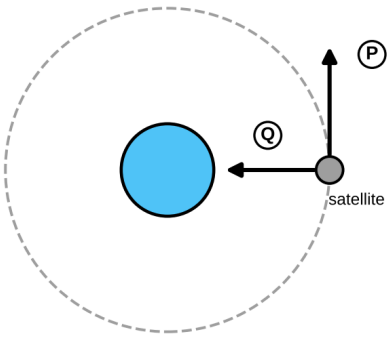
0 1 . 3 What is a **galaxy**? **[1 mark]**

0 1 . 4 Describe how the Solar System was formed. **[3 marks]**

0 1 . 5 Give **two** differences between a planet and a moon. **[2 marks]**

0 1 . 6 Explain how the Sun produces energy. **[2 marks]**

0 2 The diagram shows a satellite in a circular orbit around the Earth.



0 2 . 1 Which arrow, **P** or **Q**, shows the direction of the gravitational force on the satellite? **[1 mark]**

0 2 . 2 The speed of the satellite is constant but its velocity is changing. Explain why. **[3 marks]**

0 2 . 3 Explain why the force of gravity does not make the satellite fall to Earth. **[2 marks]**

0 2 . 4 A satellite moves into a higher orbit. State what happens to its orbital speed and its time period. **[2 marks]**

0 2 . 5 Write down the equation that links orbital speed, orbital radius and time period. **[1 mark]**

0 2 . 6 The International Space Station orbits at a radius of 6.8×10^6 m with a speed of 7700 m/s. Calculate the time period in minutes. **[4 marks]**

Answer = _____ minutes

0 2 . 7 The Earth orbits the Sun at a radius of 1.5×10^{11} m with a time period of 3.2×10^7 s. Calculate the orbital speed of the Earth. **[4 marks]**

Answer = _____ m/s

0 3 Satellites are used for communications and for monitoring the Earth.

0 3 . 1 Describe what is meant by a **geostationary** orbit. **[3 marks]**

0 3 . 2 Give **one** use of a geostationary satellite. **[1 mark]**

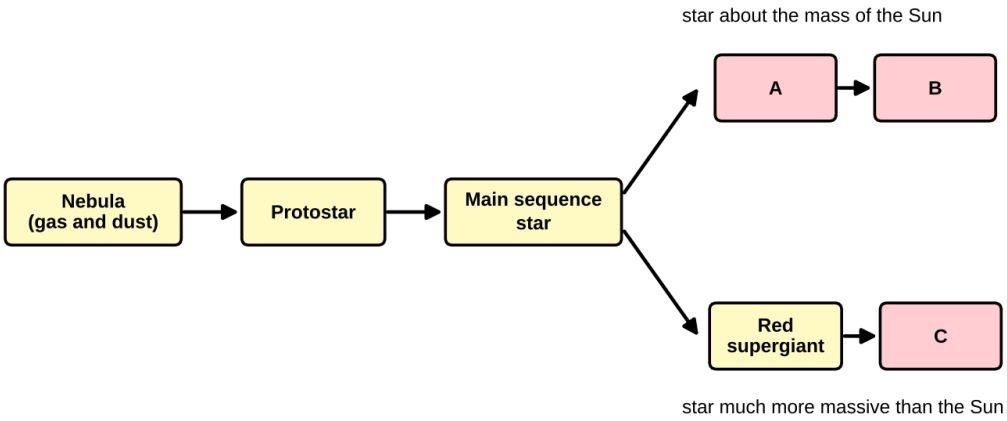
0 3 . 3 A weather-monitoring satellite is placed in a low polar orbit. Explain one advantage of this orbit. **[2 marks]**

0 3 . 4 Give **one** reason why satellites have solar panels. **[1 mark]**

0 3 . 5 The Moon is a natural satellite of the Earth. State the force that keeps it in orbit. **[1 mark]**

0 3 . 6 A comet orbits the Sun in an elliptical orbit. Explain why the comet travels fastest when it is nearest the Sun. **[3 marks]**

0 4 The diagram shows the life cycle of two different types of star.



0 4 . 1 Name the stages **A**, **B** and **C**. **[3 marks]**

0 4 . 2 State what forms after the supernova stage in a star much more massive than the Sun. **[2 marks]**

0 4 . 3 State the stage after a white dwarf cools down. **[1 mark]**

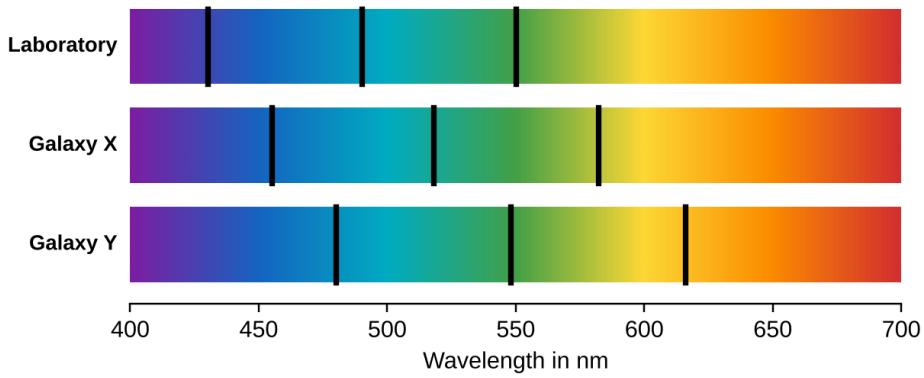
0 4 . 4 Describe how a protostar becomes a main sequence star. **[3 marks]**

0 4 . 5 Explain why a main sequence star is stable. **[2 marks]**

0 4 . 6 Explain how the elements heavier than iron are formed. **[2 marks]**

0 4 . 7 Explain why the Sun will eventually become a red giant. **[2 marks]**

0 5 The diagram shows the absorption lines in the spectrum of light from a laboratory source and from two distant galaxies, **X** and **Y**.



0 5 . 1 Describe how the lines for galaxy **X** differ from the laboratory lines. **[2 marks]**

0 5 . 2 What name is given to this effect? **[1 mark]**

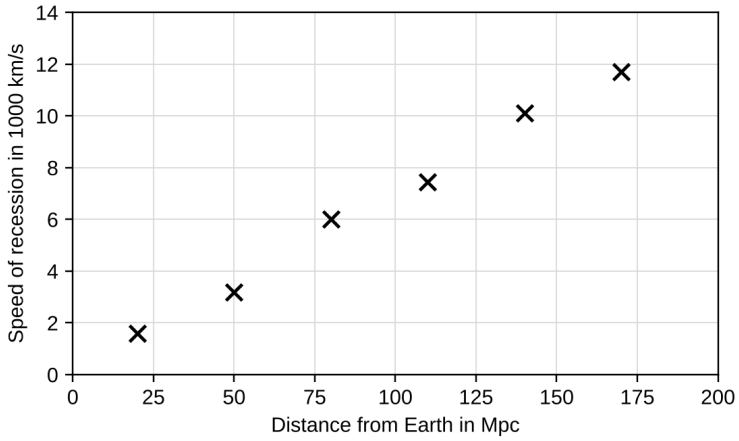
0 5 . 3 What does this effect tell scientists about the movement of the galaxies? **[1 mark]**

0 5 . 4 Galaxy **Y** has the greater shift. Explain what this shows about galaxy **Y**. **[2 marks]**

0 5 . 5 Use the diagram to find the shift in wavelength of the line at 430 nm in the laboratory spectrum for galaxy **Y** (480 nm). **[2 marks]**

0 5 . 6 Explain why the observed wavelength of light from a source moving away from an observer increases. **[2 marks]**

0 6 The graph shows how the speed at which galaxies move away from Earth depends on their distance from Earth.



0 6 . 1 Draw a line of best fit through the points and the origin. **[1 mark]**

0 6 . 2 Describe the relationship between the distance of a galaxy and its speed of recession. **[2 marks]**

0 6 . 3 Use the graph to find the speed of a galaxy at a distance of 100 Mpc. **[2 marks]**

0 6 . 4 Explain what this evidence suggests about the universe. **[2 marks]**

0 6 . 5 Describe the Big Bang theory. **[3 marks]**

0 6 . 6 Give **one** piece of evidence, other than red-shift, that supports the Big Bang theory. **[1 mark]**

0 6 . 7 Dark matter and dark energy are used to explain some observations. State what is meant by dark matter. **[2 marks]**

0 7 This question is about scientific ideas about the Universe.

0 7 . 1 Scientists once believed in the 'Steady State' theory, in which the Universe has always existed. Explain why the discovery of the cosmic microwave background radiation supported the Big Bang theory rather than the Steady State theory. **[3 marks]**

0 7 . 2 Suggest why scientists need new observations to confirm theories about the early Universe. **[2 marks]**

0 7 . 3 Explain why light from distant galaxies can be used to look back in time. **[2 marks]**

0 7 . 4 A galaxy is 100 million light-years away. State what a light-year measures. **[1 mark]**

0 8 Read the question and answer in full sentences.

0 8 . 1 *Describe the life cycle of a star that is about the same mass as the Sun. Include the forces acting on the star at each stage. **[6 marks]**

0 8 . 2 *Explain what red-shift tells us about the Universe, and how it provides evidence for the Big Bang theory. **[6 marks]**

END OF QUESTIONS

MARK SCHEME – Topic 4.8: Space physics

How to use this mark scheme: mark your own work in a different colour. A semicolon (;) separates marking points and each is worth one mark unless stated. A slash (/) separates alternative answers that are equally acceptable. Words in brackets are not needed to gain the mark. **Allow** = acceptable; **Ignore** = neither credited nor penalised; **Do not accept** = contradicts or spoils the answer. Marks are for the *science*, not the wording.

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.1	P: (nearest) planet / Mercury ; Q: (Earth-like) planet with a moon ; R: asteroid belt ; S: outer (gas giant) planet ;	The Sun is at the centre. Accept planet names Mercury / Earth / Jupiter or equivalent.	4	AO1 4.8.1.1
01.2	Milky Way		1	AO1 4.8.1.1
01.3	a very large group of stars (held together by gravity)		1	AO1 4.8.1.1
01.4	a cloud of dust and gas (nebula) ; was pulled together by gravity ; forming the Sun at the centre and planets from the remaining material ;		3	AO1 4.8.1.1
01.5	planets orbit a star; moons orbit planets ; planets are larger ; moons are natural satellites ;	Any two.	2	AO1 4.8.1.1
01.6	nuclear fusion ; of hydrogen nuclei to form helium ;		2	AO1 4.8.2.1
02.1	Q		1	AO2 4.8.1.2
02.2	velocity is a vector (has speed and direction) ; direction of motion changes continuously ; so velocity changes (the satellite is accelerating towards the Earth) ;		3	AO2 4.8.1.2
02.3	the satellite has a high speed (tangential velocity) ; gravity changes its direction but it keeps 'falling around' the Earth ;		2	AO3 4.8.1.2
02.4	speed decreases ; time period increases ;		2	AO2 4.8.1.2
02.5	orbital speed = (2 π × orbital radius) ÷ time period		1	AO1 4.8.1.2
02.6	T = 2π r ÷ v ; = 2 × π × 6.8 × 10^6 ÷ 7700 ; 5550 (s) ; 92 (minutes) ;	Allow 92–93 minutes.	4	AO3 4.8.1.2
02.7	v = 2π r ÷ T ; = 2π × 1.5 × 10^{11} ÷ 3.2 × 10^7 ; 29 450 (m/s) ; 2.9 × 10^4 (m/s) ;	Allow 3.0 × 10^4.	4	AO3 4.8.1.2
03.1	period of 24 hours ; above the equator ; satellite stays above the same point on the Earth ;		3	AO1 4.8.1.2

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.2	satellite TV / communications / weather		1	AO1 4.8.1.2
03.3	it passes over the whole Earth as the Earth rotates ; closer to the Earth so it gives more detailed images ;		2	AO2 4.8.1.2
03.4	to supply electrical energy (no fuel / long-lasting)		1	AO1 4.8.1.2
03.5	gravity / gravitational force (attraction) from the Earth		1	AO1 4.8.1.2
03.6	gravitational force is greatest nearest the Sun ; work done by gravity transfers energy from gravitational potential to kinetic ; so the comet speeds up ;		3	AO3 4.8.1.2
04.1	A: red giant ; B: white dwarf ; C: supernova ;		3	AO1 4.8.2.1
04.2	neutron star ; or black hole (very massive stars) ;		2	AO1 4.8.2.1
04.3	black dwarf		1	AO1 4.8.2.1
04.4	gravity pulls dust and gas together, temperature rises ; when hot enough, hydrogen nuclei fuse (nuclear fusion) ; outward pressure from fusion balances gravity (stable star) ;		3	AO1 4.8.2.1
04.5	inward gravitational force ; balanced by the outward force (pressure) from radiation / fusion energy ;		2	AO2 4.8.2.1
04.6	in a supernova / explosion of a massive star ; extreme conditions fuse nuclei to form heavier elements ;		2	AO2 4.8.2.1
04.7	the Sun runs out of hydrogen in its core ; fusion of helium / outer layers expand and cool ;		2	AO2 4.8.2.1
05.1	shifted towards the red end ; longer wavelengths / lower frequency ;		2	AO3 4.8.3.1
05.2	red-shift		1	AO1 4.8.3.1
05.3	they are moving away (from the Earth)		1	AO2 4.8.3.1
05.4	moving away faster ; (and so is probably further away) ;		2	AO3 4.8.3.1
05.5	480 − 430 ; 50 (nm) ;		2	AO3 WS 3.1
05.6	the wave crests are stretched out / source moves away between wave emissions ; wavelength is increased (Doppler effect) ;		2	AO2 4.8.3.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.1	straight line through the origin and the points ;		1	AO3 WS 3.3
06.2	the further away, the faster it moves away ; directly proportional ;		2	AO3 4.8.3.1
06.3	reading from the line at 100 ; 7000 km/s (6500–7500) ;		2	AO3 WS 3.1
06.4	the universe is expanding ; all galaxies are moving apart (space itself is expanding) ;		2	AO3 4.8.3.1
06.5	the universe began from a very small / hot / dense point ; which exploded (expanded) about 13.8 billion years ago ; and has been expanding since ;		3	AO1 4.8.3.2
06.6	cosmic microwave background radiation (CMBR)		1	AO1 4.8.3.2
06.7	matter that cannot be seen / detected directly ; inferred from gravitational effects on galaxies' rotation ;		2	AO1 4.8.3.2
07.1	Big Bang predicts leftover radiation from the early hot universe ; this radiation has been stretched to microwave wavelengths as the universe expanded ; Steady State cannot explain this uniform background ;		3	AO3 4.8.3.2
07.2	theories can only be accepted if consistent with evidence ; new evidence may support or disprove them (peer review) ;		2	AO3 4.8.3.2
07.3	light takes a long time to travel across space ; so we see the galaxy as it was when the light was emitted ;		2	AO2 4.8.3.1
07.4	distance (that light travels in one year)		1	AO1 4.8.3.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
08.1	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • Level 3 (5–6): all stages from nebula to black dwarf in order, with forces / fusion / balance explained correctly. • Level 2 (3–4): most stages with some reference to forces. • Level 1 (1–2): simple statements. • Indicative content: nebula (gas and dust) pulled together by gravity → protostar heated as it contracts → fusion starts, main sequence, gravity balanced by radiation pressure → hydrogen runs out: star expands and cools, red giant → outer layers drift away, core contracts, white dwarf (gravity compresses the core) → cools, fades to a black dwarf.		6	AO1 4.8.2.1
08.2	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • Level 3 (5–6): clear link between red-shift, recession speed vs distance, expansion and a beginning from a small point. • Level 2 (3–4): some correct explanation. • Level 1 (1–2): simple statements. • Indicative content: light from distant galaxies shifted to longer wavelength (red-shift); galaxies are moving away; further galaxies more red-shifted so move faster; the universe is expanding; running time backwards means everything was once close together; Big Bang; supported by CMBR; dark matter / energy still unexplained.		6	AO2 4.8.3

Where marks are lost in Topic 4.8

Use this page after marking. Tick the ones that cost you marks, then redo those questions closed-book.

- ☐ **Life cycle:** learn the two routes — Sun-like: nebula → protostar → main sequence → red giant → white dwarf → black dwarf; massive: … → red supergiant → supernova → neutron star or black hole.
- ☐ **Stable star** = gravity (inwards) balanced by radiation / fusion pressure (outwards). Say **both** forces.
- ☐ **Orbits:** speed is constant but velocity changes because direction changes. Gravity is the centripetal force. Higher orbit → slower and longer period.
- ☐ **$v = 2\pi r \div T$:** radius in metres, time in seconds. Convert minutes/hours/days first.
- ☐ **Red-shift** = light stretched to longer wavelength because the source is moving away. More red-shift = faster and (usually) further away.
- ☐ **Evidence for the Big Bang:** red-shift (expanding universe) and cosmic microwave background radiation. Dark matter and dark energy are still unexplained.
- ☐ **Fusion** in stars makes elements up to iron; heavier elements form in supernovae.